

Cancer Surveillance: From Data to Public Health Action

Asynchronous Session: Concepts, Practice & Application

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Course:

PUBLIC HEALTH TEACHING AND TRAINING



Learning Objectives



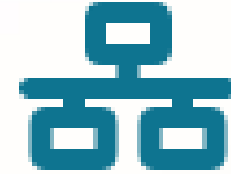
Define & Compare

Distinguish between passive, active, sentinel, and syndromic surveillance methods.



Apply Rules

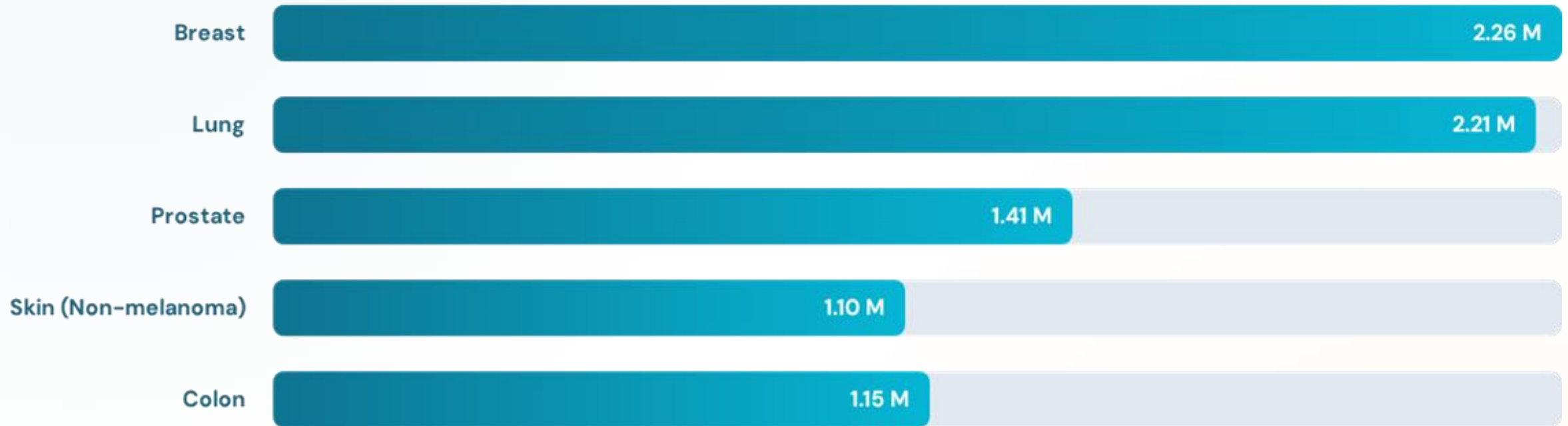
Utilize ICD-O-3, staging systems, and reportability rules to classify cases accurately.



Identify Workflow

Map the end-to-end data lifecycle from case finding to quality reporting.

Global Cancer Burden: Incidence (2022)



Breast and Lung cancer account for the highest volume of new cases globally.

Global Cancer Burden: Mortality (2022)



Lung cancer dominates mortality, highlighting the need for early detection.

The Burden Summary

Incidence Insights

- ~20 million new cases annually.
- Breast cancer is the most diagnosed.
- High prevalence implies a growing need for survivorship care.

Mortality Insights

- ~9.7 million deaths annually.
- Lung cancer is the leading cause of death.
- Disparities in outcomes persist globally.

Surveillance Approaches

Type	Mechanism	Best For	Challenges
 Passive	Facilities submit reports	 Scale & Coverage	 Variable Quality
 Active	Registry staff seek data	 Completeness	 Resource Intensive
 Syndromic	Selected sites	 Rare Cancers	 Representativeness
 Syndromic	Early signals (ED/Rx)	 Early Warning	 Low Specificity

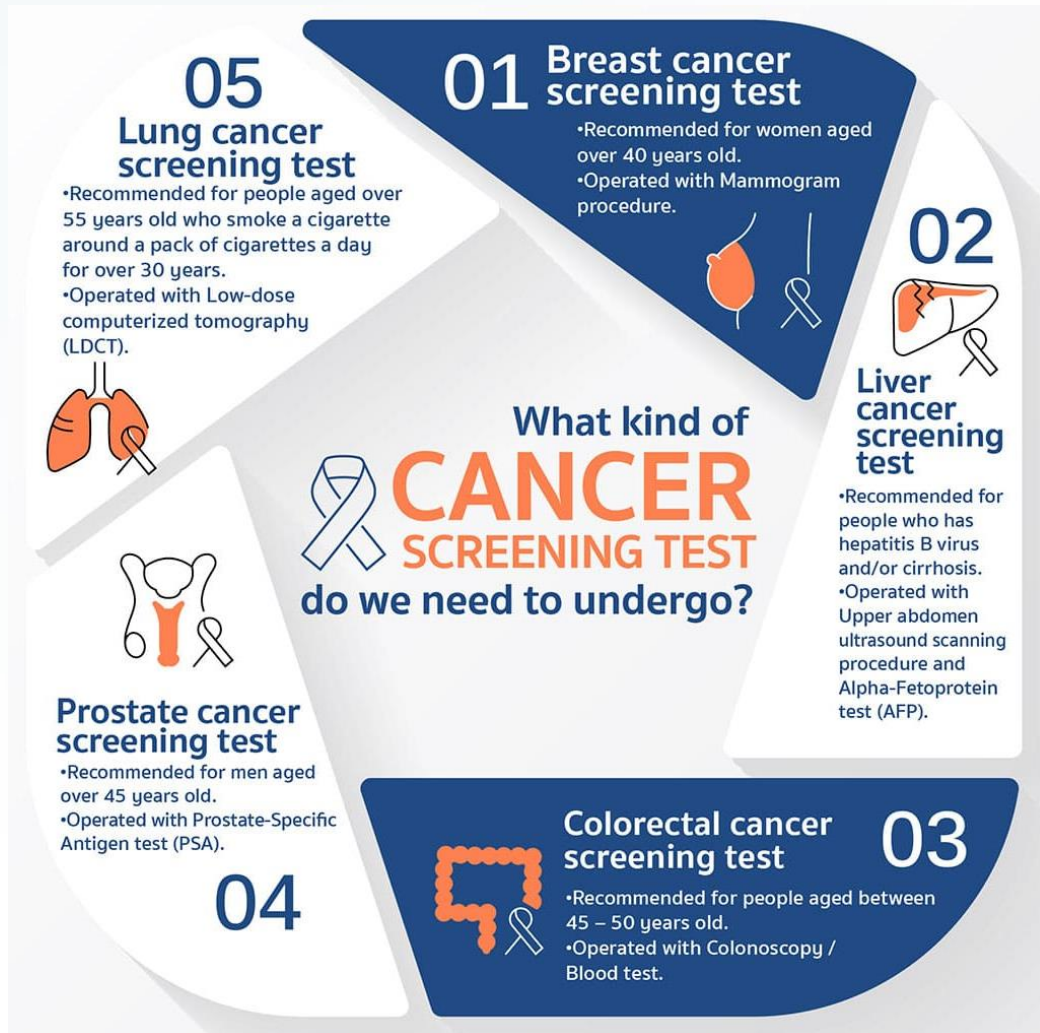
From Data to Screening

- 🫁 **Lung:** Low-dose CT for high-risk smokers.
- 👙 **Breast:** Mammograms for women 40+.
- 🔍 **Colorectal:** Colonoscopy or Stool tests for 45+.
- ♂ **Prostate:** PSA testing for men 45+.



Effective surveillance identifies at-risk populations to target these interventions.

Right Data → Right Test → Right Time



Surveillance turns signals into saved lives.

Defining the Case

ICD-O-3 Standards

The global standard for coding cancer cases.

- **Topography:** Site of origin (e.g., C50.9 Breast)
- **Morphology:** Cell type (e.g., 8500/3 Infiltrating Duct)

Reportability Rules

Determines which cases must be included in the registry.

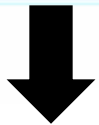
- **Malignant (/3):** Always reportable.
- **In Situ (/2):** Most are reportable
- **Benign (/0):** Generally not, except CNS.

Staging & Rules



TNM Staging

- Tumor size
- Nodes involved
- Metastasis.



**Used for clinical
treatment planning**



SEER Summary

- Localized
- Regional
- Distant.



**Used for population-level
comparison and Epidemiology**



Date Rules

Anchor to the first
definitive diagnosis
(usually the pathology
report date).

The Data Ecosystem

Cancer data is an aggregation of multiple streams:



Pathology

The gold standard for diagnosis.



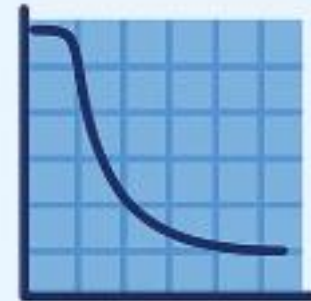
EHRs

Clinical context and history.



Claims

Treatment verification.



Vital Records

Survival outcomes.

5 Dimensions of Data Quality



COMPLETENESS

Are we capturing all eligible cases?



CONSISTENCY

Are standards applied uniformly over time?



ACCURACY

Is the data correct and valid?



TIMELINESS

Is data available rapidly enough for action?



UNIQUENESS

Are there duplicates in the System?

NPCR Data Quality Targets



Meeting these standards ensures data is “Fit for Use”

Privacy & Security Cycle

Protecting Patient Trust



RBAC: Role-Based Access Control ensures only authorized staff see data



Encryption: Data is encrypted in transit (VPN) and at rest.



Audit Logs: Every access is tracked and monitored.

Reporting Safely

Balancing usefulness and privacy in Cancer data



Small-Cell Suppression

- Never publish counts fewer than <5-10 cases (follow state laws and policy).
- Combine or suppress cells to prevent re-identification in small populations



Aggregation

- Report data by larger geographic areas (e.g., region vs. zip code)
- Use multi-year aggregates (e.g., 3-5-years) to ensure statistical stability and privacy.

Goal: useful data, low disclosure risk

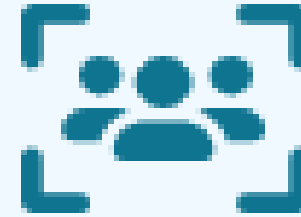


Session Resources



Skeleton Notes

- A fillable handout to help you follow along and capture key metrics during the session.
- Check the chat for the download link.



Case Study Guide

A reference sheet for the "**Rural Spike**" breakout activity, including the rubric for your group's diagnosis.

Questions?

Thank you for your attention

Next Steps: Complete the post-session assessment.